scientific reports

Check for updates

OPEN

Probabilistic machine learning for noisy labels in Earth observation

Spyros Kondylatos[1,2]✉, Nikolaos Ioannis Bountos[1,3], Ioannis Prapas[1,2], Angelos Zavras[1,3], Gustau Camps-Valls[2] & Ioannis Papoutsis[1,4]

Label noise poses a significant challenge in Earth Observation (EO), often degrading the performance and reliability of supervised Machine Learning (ML) models. Yet, given the critical nature of several EO applications, developing robust and trustworthy ML solutions is essential. In this study, we take a step in this direction by leveraging probabilistic ML to model input-dependent label noise and quantify data uncertainty in EO tasks, accounting for the unique noise sources inherent in the domain. We train uncertainty-aware probabilistic models across a broad range of high-impact EO applications—spanning diverse noise sources, input modalities, and ML configurations—and introduce a dedicated pipeline to assess their accuracy and reliability. Our experimental results show that the uncertainty-aware models outperform standard deterministic approaches across most datasets and evaluation metrics. Moreover, through rigorous uncertainty evaluation, we validate the reliability of the predicted uncertainty estimates, enhancing the interpretability of model predictions. Our findings emphasize the importance of modeling label noise and incorporating uncertainty quantification in EO, paving the way for more reliable solutions in the field.

The increasing availability of Remote Sensing (RS) data has significantly advanced research and applications across various scientific fields[1]. Satellite-based Earth Observation (EO) has transformed the analysis and modeling of the Earth system, enabling continuous monitoring of environmental, natural, and human-driven processes of the globe[2]. By providing large-scale data from space, EO offers critical insights into the dynamics and interactions of the Earth's five major spheres—Atmosphere, Hydrosphere, Geosphere, Biosphere, and Cryosphere. This wealth of data supports a wide array of applications, such as climate change[3], food insecurity[4], ecosystem monitoring[5], and Earth Science applications[6], fostering a deeper understanding of the complex systems that shape our planet.

Deep Learning (DL) has emerged as a powerful tool for extracting knowledge from EO data. DL methods have been successfully applied across a wide range of EO modalities, including optical imagery, Synthetic Aperture Radar (SAR), Interferometric Synthetic Aperture Radar (InSAR), and multi-modal data[7]. Supervised DL models, trained on annotated datasets from these sources, have demonstrated remarkable performance in addressing several EO and geoscientific problems[8,9], showcasing great results across different applications[10,11]. However, despite their widespread adoption within the EO community, the effectiveness of supervised DL approaches remains highly dependent on the availability of large, high-quality labeled datasets[12].

Annotating EO data presents unique challenges, often requiring expert domain knowledge tailored to specific tasks and applications; yet, manual annotation of large-scale datasets by domain experts is both costly and time-consuming. To overcome these limitations, alternative strategies such as crowdsourcing, semi-supervised learning, or annotations by ML practitioners are frequently used[13]. While these methods improve scalability and reduce costs, they also introduce a higher risk of labeling errors, potentially compromising the quality of supervised datasets. Even when expert annotations are employed, the process remains complex due to the intricate nature of EO applications. These applications typically require a deep understanding of dynamic Earth processes, as well as the ability to disentangle complex signals and interpret natural phenomena from satellite imagery. As a result, the label generation process becomes uncertain[14], often resulting in label noise within EO supervised datasets. In single-label classification, label noise occurs when a sample is given the wrong label. In multi-label classification, it can appear as either missing labels, where a relevant label is not assigned to a sample, or incorrect labels, where a sample is mistakenly linked to a label that does not actually apply. Label noise can also result from a combination of both missing and incorrect labels[13]. Addressing label noise is crucial, as its presence can degrade the performance of supervised models, ultimately leading to less reliable predictions[15].

Reliability is a fundamental requirement for EO DL models[16,17]. This is particularly relevant for EO applications related to disaster management, such as wildfire forecasting[18], flood mapping[19], landslide

[1]Orion Lab, National Observatory of Athens & National Technical University of Athens, 15772 Athens, Greece. [2]Image Processing Laboratory (IPL), Universitat de València, 46980 Valencia, Spain. [3]Harokopio University of Athens, 17676 Athens, Greece. [4]Archimedes, Athena Research Center, 15125 Athens, Greece. ✉email: skondylatos@noa.gr

monitoring[20], and volcanic unrest detection[21]. These applications demand swift and precise decision-making, as poor choices can lead to loss of lives or environmental damage. A key approach for improving the reliability of DL models relies on quantifying the uncertainty in their predictions. Assessing the confidence level of model outputs can enhance decision-making, allowing stakeholders to adapt their strategies accordingly. For instance, in volcanic activity early-warning systems, emergency response protocols may vary depending on the model's confidence. A highly certain prediction of an imminent eruption may initiate large-scale evacuations and resource allocation. Conversely, a high-uncertainty prediction may lead decision-makers to opt for closer monitoring and precautionary measures rather than immediate, large-scale interventions.

In Machine Learning (ML), uncertainty is typically classified into two main types: epistemic (model) uncertainty and aleatoric (data) uncertainty. Epistemic uncertainty arises from a model's lack of knowledge and can be reduced as more data become available or the model improves. On the other hand, aleatoric uncertainty stems from inherent noise in the data and cannot be reduced, even with additional data samples. Aleatoric uncertainty can be characterized as homoscedastic (input-independent), where the noise remains constant across all samples in the input space, and heteroscedastic (input-dependent), where the level of noise varies depending on the input[22,23].

The relationship between heteroscedastic label noise and aleatoric uncertainty is significant as the latter increases in samples that are more challenging to annotate. This connection has been explored within the Computer Vision (CV) community, leading to the development of probabilistic models that estimate heteroscedastic aleatoric uncertainty by modeling input-dependent label noise[23,24]. In these models, a distribution is typically placed over the logits of a Neural Network (NN), where the mean represents the predicted output and the variance quantifies the associated uncertainty. These methods have improved model performance and enabled the quantification of data uncertainty across various CV tasks impacted by label noise. To the best of our knowledge, such methods have yet to be explored in the EO domain.

The widespread presence of label noise, coupled with the growing need for reliable uncertainty estimation in EO tasks, highlights the potential of such methods in EO. In this work, we adopt the probabilistic framework introduced by Collier et al.[24], to model input-dependent label noise and quantify data uncertainty in EO tasks. To systematically assess the framework's effectiveness in handling the diverse sources of label noise specific to the domain, we apply it to four EO applications, each characterized by a distinct source of label noise, varying input modalities, and ML setups. We assess its predictive performance against standard deterministic DL models and leverage its probabilistic nature to quantify estimates of the aleatoric uncertainty, while validating the lack of the epistemic signal. To examine the practical utility of these uncertainties, we use two complementary approaches: (i) we assess their reliability through dedicated evaluation strategies, and (ii) we use the predicted uncertainty estimates to identify certain and uncertain samples across the various EO tasks, demonstrating their actual applicability in real-world applications.

The main contributions of this work can be summarized as follows:

- We provide a comprehensive classification of label noise sources specific to EO datasets.
- We investigate the application of probabilistic ML to mitigate the impact of label noise in EO, demonstrating its effectiveness across four high-stakes applications.
- We propose a dedicated pipeline for integrating uncertainty-aware ML in EO, incorporating uncertainty quantification, evaluation, and visualization. This framework highlights the critical role of uncertainty in enhancing model reliability and supporting informed decision-making in critical EO tasks.

Background & related work

Learning under label noise in deep learning

In supervised DL, a finite set of training data denoted as:

$$D = \{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\} = X \times Y$$

is used for training, with X representing the input data and Y the corresponding labels. A noisy training dataset is typically defined as:

$$D = X \times \tilde{Y},$$

where $\tilde{Y}$ represents the set of observed labels that may differ from the true labels Y. Various methods have been proposed to mitigate the effects of label noise in DL[25]. Regularization strategies reduce the memorization of noisy labels by modifying the loss function[26–28]. Additionally, loss correction techniques[29–31] and reweighting[32–34] adjust the loss dynamically to alleviate the impact of label noise. Other methods focus on identifying potentially mislabeled samples, which are downweighted or discarded during training[35]. MentorNet[15] uses an auxiliary network to assign weights for potentially noisy examples, while MentorMix[36] enhances this with mixup regularization[37]. Co-teaching[38] trains two neural networks simultaneously, where small loss samples from one network are fed to the other network for learning. Despite the variety of existing methods, EO applications require not only robustness but also reliability. In this context, uncertainty estimation is essential, and this work focuses on approaches that simultaneously address label noise and quantify uncertainty to ensure trustworthy modeling.

Uncertainty in supervised deep learning

Considering the set of training data:

$$D = X \times Y,$$

a DL model f^w, parameterized by weights w is typically trained to map $f^w(x) = y$ for each pair (x, y) in D, without accounting for the inherent uncertainty both in the data and the model itself. To address this limitation, predictive variance networks have been introduced to estimate total predictive uncertainty in regression neural networks[39]. Several approaches have also been proposed to improve uncertainty calibration in DL models[40,41]. Moreover, conformal prediction has emerged as a statistically rigorous framework for generating uncertainty estimates[42]. To distinguish between different sources of uncertainty, Prior Networks have been proposed to disentangle epistemic and aleatoric uncertainty[43]. Additional methods focus on estimating either epistemic or aleatoric uncertainty individually[16].

Epistemic uncertainty is commonly captured through Bayesian Neural Networks (BNNs)[44,45] and Deep Ensembles (DEs)[46]. Aleatoric uncertainty is calculated using: (i) external methods, applied to an already trained deterministic NN[47–49], (ii) network modifications, where the architecture is adapted to predict uncertainty during training, often by learning the parameters of a probabilistic distribution in the output space[50,51], and (iii) test-time data augmentation, where the model generates multiple predictions by augmenting inputs at inference time and calculate uncertainty as their variability[52,53].

Kendall and Gal[23] introduced a network modification method for quantifying heteroscedastic aleatoric uncertainty by explicitly modeling label noise. In this method, a Gaussian distribution is placed on the logits of a standard softmax classification model before applying the softmax function. Formally, given an input x, the logits $u_c(x)$ are modeled as:

$$u_c(x) \sim \mathcal{N}(f_c^w(x), \sigma_c^w(x)^2), \forall c = 1, \ldots, K,$$

where $f_c^w(x)$ represents the mean logit prediction and $\sigma_c^w(x)^2$ the variance, both of which are learned by the model. The final class probabilities are then obtained via the softmax function:

$$p_c(x) = \frac{\exp\left(u_c(x)\right)}{\sum_{k=1}^{K} \exp(u_k(x))}, \forall c = 1, \ldots, K. \tag{1}$$

where K is the total number of classes. In this setting, the variance term $\sigma_c^w(x)^2$ allows the model to quantify aleatoric uncertainty.

Uncertainty & noisy labels in remote sensing and geoscience

Gaussian Processes[54], and Deep Gaussian Processes[55] have been applied in several EO tasks[56,57], such as biophysical parameter retrieval[58,59] and RS image classification[60]. While these models offer quality estimates of uncertainty, their complexity poses challenges in training, making their utilization in big Earth data problems computationally infeasible. Lately, more scalable BNNs have been proposed for uncertainty quantification in EO. In particular, Monte Carlo (MC) Dropout[61] has been employed for biophysical parameter retrieval[62], ice and water detection[63], urban landscape object segmentation[64], and aerial image segmentation[65]. Moreover, Bayesian Deep Learning[66] and VI-based Bayesian Convolutional Neural Networks (CNNs)[67] have been used for hyperspectral image classification.

Additional uncertainty-aware applications include corn yield prediction[68], soil moisture retrieval[69], crack segmentation under noisy drone annotations[70], and nitrogen dioxide estimation[71]. Singh et al.[72] have highlighted conformal prediction as a promising direction for uncertainty quantification in EO. Moreover, evidential DL with Dirichlet distributions has been proposed for drought prediction[73], and other Earth system science applications[74]. In landslide modeling, MC simulations have been used to quantify uncertainty in displacement forecasting[75], while entropy-based uncertainty methods have improved DL model reliability for drought detection[76].

Furthermore, probabilistic generative models have shown promise in Geoscience. Diffusion models have been used for wildfire spread prediction[77], sea-ice surrogate modeling[78], and meteorological state resolution enhancement[79], while Variational Autoencoders have supported unsupervised anomaly detection in avalanche[80] and flood detection[81].

Addressing label noise in RS has been the focus of several studies. Heteroscedastic Gaussian Processes have been used for biophysical parameter retrieval in cases where noise is signal-dependent[82]. Additionally, task-specific methodologies have been explored. Li et al.[83] proposed a multi-view CNN framework that iteratively corrects label errors, while Jiang et al.[84] introduced a random label propagation algorithm to cleanse label noise for hyperspectral image classification. To enhance model resilience, a categorical cross-entropy loss function has been employed to train CNNs robustly for RS image classification[85]. Xu et al.[86] proposed a dual-channel residual network combined with a noise-robust loss function to reduce the influence of mislabeled samples. Noisy-tolerant methods have also been used for high-impact applications like drought detection[87], where a label correction method relying on model outputs was used. Other strategies include complementary learning and deep metric learning with a robust softmax loss to mitigate the adverse effects of noisy labels[88]. Methods for addressing multi-label noise in RS include collaborative learning approaches[89], generative reasoning strategies[13], and pseudo-labeling[90].

Method

The relationship between label noise and the inherent aleatoric uncertainty remains largely unexplored in the EO field. In this work, we take a step toward addressing this gap. We use the probabilistic framework introduced by

Collier et al.[24] to model heteroscedastic label noise and quantify aleatoric uncertainty in EO tasks. The method builds upon the standard heteroscedastic classification model of Kendall and Gal[23] (see Eq. 1). We provide a brief overview of the method for completeness and refer readers to the original work for more details.

Probabilistic ML framework

The method considers a generative process for the labels. This is handled by a latent variable $u_c(x)$, which is associated with each class c and input x. This variable is the sum of a deterministic vector $f_c^w(x)$ and an unobserved stochastic component ϵ_c, expressed as: $u_c(x) = f_c^w(x) + \epsilon_c$. A label is generated by sampling from $u_c(x)$ and taking the arg max of all classes, i.e. class c^* is the generated label if $u_c(x) \leq u_{c*}(x), \forall c \in 1, \dots K$, where K is the total number of classes. The probability $p_c(x)$ that an input x belongs to class c, can be then expressed as

$$\begin{aligned} p_c(x) &= P(\arg\max_k u_k(x) = c) \\ &= \int \mathbf{1}\left\{\arg\max_k u_k(x) = c\right\} p(\epsilon_c)\, d\epsilon_c \end{aligned} \tag{2}$$

Assuming that ϵ_c are independent and identically distributed (homoscedastic), after calculations, the probability $p_c(x)$ becomes the well-known softmax function used in standard ML classification:

$$p_c(x) = \frac{\exp u_c(x)}{\sum_{k=1}^{K} \exp(u_k(x))}$$

Thus, this generative process is implicitly assumed in the training of standard ML classifiers. However, under the presence of input-dependent label noise, the assumption of identically distributed ϵ_c becomes restrictive, as the noise source varies from sample to sample and across different classes. In such cases, it is necessary to account for differing levels of stochasticity for each sample (heteroscedasticity). For instance, in land-use classification, images containing a mix of agricultural and arable land often present higher labeling uncertainty than more clear classes, such as forests.

To handle this heteroscedasticity, the method introduces a dependency between the noise terms ϵ_c, the input, and classes, breaking the identically distributed assumption. Specifically, $\epsilon_c \sim \mathcal{N}(0, \sigma_c^w(x)^2)$, where $\mathcal{N}$ denotes a Normal distribution, and $\sigma_c^w(x)^2$ models noise levels that vary based on the input and class. Under these assumptions, computing $p_c(x)$ in Eq. 2 is intractable. However, it can be approximated using a temperature-scaled softmax and MC sampling. Moreover, to enable gradient-based optimization, $u_c(x)$ can be reparameterized as $u_c(x) = f_c^w(x) + \sigma_c^w(x)\mu_c$, where $f_c^w(x)$ and $\sigma_c^w(x)$ are deterministic components and $\mu_c \sim \mathcal{N}(0, 1)$.

The final calculation of $p_c(x)$ is obtained as:

$$\begin{aligned} p_c(x) &= P(\arg\max_k u_k(x) = c) \\ &\approx \mathbb{E}_{\epsilon_k \sim \mathcal{N}(0, \sigma_c^w(x)^2)} \left[\frac{\exp\left(u_c(x)/\tau\right)}{\sum_{k=1}^{K} \exp\left(u_k(x)/\tau\right)} \right], \tau > 0 \\ &\approx \frac{1}{S} \sum_{s=1}^{S} \frac{\exp((f_c^w(x) + \sigma_c^w(x)\mu_c^s)/\tau)}{\sum_{k=1}^{K} \exp((f_k^w(x) + \sigma_k^w(x)\mu_k^s)/\tau)}, \ \mu_c^s \sim \mathcal{N}(0, 1), \end{aligned} \tag{3}$$

where S is the number of MC samples.

In this setting, the variance term $\sigma_c^w(x)^2$ is critical allowing the model to: (i) discern label noise varying across individual samples, learning to assign higher values to inputs exhibiting higher noise and lower values to those with less noise; and (ii) provide aleatoric uncertainty for the predictions of each sample. The practical implementation of this solution is simple. The only modification relative to standard neural networks is limited to the final layer of the network, which is adapted to output twice the usual number of outputs, providing both the mean $f_c^w(x)$ and the variance $\sigma_c^w(x)^2$ of the induced Normal distribution for each class. MC sampling is used to generate multiple samples from this distribution, allowing the computation of the final predictions as outlined in Eq. (3). This operation is computationally efficient because the sampling is restricted to the final layer of the network, unlike, for instance, BNNs, which involve sampling across all network parameters. The temperature parameter τ plays a critical role in balancing the trade-off between the approximation bias and variance in the MC gradient estimates. Its optimal value requires separate computation for each task using a validation set. Notably, when $\tau = 1$, the method converges to the standard heteroscedastic classification model (Eq. (1)).

Aleatoric uncertainty estimation

As previously discussed, the term $\sigma_c^w(x)^2$ enables the estimation of aleatoric uncertainty for the predictions. In this study, to quantify this uncertainty, we use the MC samples from the predictive distribution in Eq. (3) and compute their variance. Specifically, for a given input x, an individual MC sample is expressed as:

$$p_{c,s} = \frac{\exp((f_c^w(x) + \sigma_c^w(x)\mu_c^s)/\tau)}{\sum_{k=1}^{K} \exp((f_k^w(x) + \sigma_k^w(x)\mu_k^s)/\tau)}$$

and the mean prediction across MC samples as:

$$\bar{p_c} = \frac{1}{S}\sum_{s=1}^{S} p_{c,s}$$

The aleatoric uncertainty for each class is computed as the variance of all drawn MC samples:

$$uncertainty_c = \frac{1}{S}\sum_{i=1}^{S}(p_{c,i} - \bar{p_c})^2.$$

Investigation of probabilistic machine learning in Earth observation

We investigate probabilistic ML for modeling under label noise in EO tasks. We employ standard deterministic DL models and integrate the probabilistic module at the logits level (Fig. 1b(2)). MC sampling is used to generate both the model predictions and the associated uncertainty estimates. Leveraging the framework's flexibility, we develop uncertainty-aware DL models for a range of high-stakes EO applications, exhibiting variations in input modalities, ML configurations, and noise sources. These applications span single-label and multi-label image classification, as well as image segmentation tasks. To assess the effectiveness of the probabilistic model, we benchmark its performance against standard deterministic DL models and the standard heteroscedastic model. Moreover, we introduce a dedicated pipeline for uncertainty-aware ML modeling in EO. This pipeline emphasizes the assessment of uncertainty estimates in addition to traditional performance evaluation, highlighting that the trustworthiness of a model is often more important than merely improving its performance.

The key steps of our pipeline are summarized in Fig. 1: (1) we define the primary label noise sources in EO supervised datasets and use datasets with noisy labels representing each category; (2) we train uncertainty-aware DL models; (3) we evaluate model performance, validating that uncertainty-aware models maintain or improve predictive accuracy compared to standard DL models; (4) we assess the reliability of uncertainty estimates to ensure their trustworthiness in EO tasks, and (5) we use the estimated uncertainties to generate visualizations of sample data, demonstrating how these uncertainties can be interpreted to support decision-making in real-world applications.

Label noise sources in Earth observation

Annotating EO images poses challenges, as different annotation strategies introduce distinct sources of label noise. Approaches such as semi-supervised labeling, non-expert annotations, or in-field data annotation each have their limitations. Additionally, the inherent complexity of EO applications and the intrinsic ambiguity in satellite data interpretation further impact label quality. In this work, we systematically define and categorize the key Noise Sources (NS) affecting labels in EO datasets as follows:

NS1: Unreliable labeling procedures. In EO applications, labels are often generated through methods such as annotation by ML practitioners, crowdsourcing initiatives, or unsupervised/semi-supervised learning techniques. While these approaches offer a cost-effective way of acquiring labeled datasets, they are generally considered less reliable regarding label quality[91].

NS2: Challenges in signal disentanglement. Even when domain experts are involved in the annotation process, the risk of inducing noise in the labels remains considerable. This may occur due to limited information provided to the experts or the complexity of distinguishing signals in satellite imagery. Furthermore, in case multiple experts label the same sample, inconsistencies in labeling results often emerge[92].

NS3: Inherent noise due to the nature of the application. In certain EO tasks, label noise is an inherent consequence of the problem being addressed. This is particularly common in applications related to natural hazards and disaster management, where similar variable conditions may or may not lead to the occurrence of specific physical phenomena. In these cases, the occurrence or non-occurrence of an event is inherently stochastic[93], making it challenging to assign clear-cut positive or negative labels.

NS4: Information loss due to annotations misalignment. An important factor contributing to label noise involves the loss of information during the dissemination of data[84]. This occurs when expert labeling is not directly applied to the data but relies on alternative sources, such as in situ measurements or field annotations. In such cases, even if the annotation process is precise, the post-alignment between input-label pairs can introduce inconsistencies.

Datasets and applications

To evaluate the effectiveness of the probabilistic model in capturing the various sources of noise in EO datasets, we carefully select four diverse datasets, each exhibiting one distinct source of label noise (Table 1). This section provides an overview of the selected datasets and tasks considered, discussing the specific noise sources associated with each.

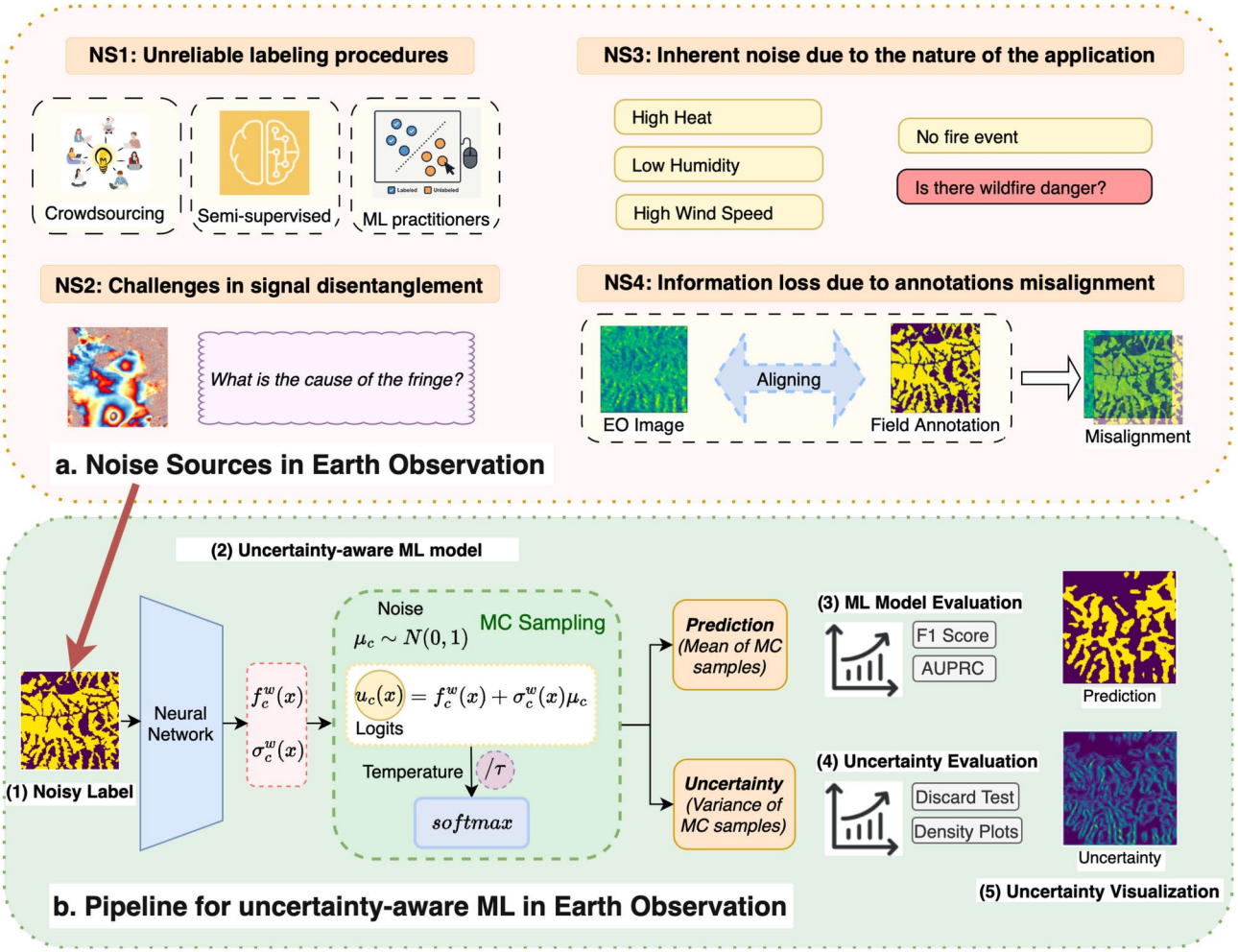


Fig. 1. (**a**) The categorization of Label Noise Sources (NS) in Earth Observation (EO), and (**b**) the uncertainty-aware Machine Learning (ML) pipeline for modeling label noise in EO proposed in this study. An uncertainty-aware ML model is used to model the label noise in EO. A Normal distribution is induced in the logits of a neural network, and the model is trained to predict its mean $f_c^w(x)$ as the output and $\sigma_c^w(x)$ as its heteroscedastic uncertainty. Monte Carlo (MC) sampling is used to generate multiple samples from this distribution, enabling the estimation of the final model prediction (mean of the samples) and its associated uncertainty (variance of the samples). A temperature parameter τ is used, scaling the logits for a tempered softmax calculation. Model performance is assessed using standard evaluation methods (F_1 score and area under precision-recall curve (AUPRC)), while uncertainty estimates are assessed using dedicated uncertainty evaluation methods (discard test and uncertainty density plots) and visualizations.

Application	Dataset	Noise source	Input modality	ML setup
Land use land cover scene classification	BigEarthNet[94]	1	Optical Images	Multi-class multi-label image classification
Landslide segmentation	Landslides dataset[20]	4	SAR	Image segmentation
Volcanic activity detection	Hephaestus[21]	2	InSAR	Image classification
Wildfire danger forecasting	Wildfires dataset[18]	3	Multi-modal	Time-series classification

Table 1. Overview of the applications, datasets, noise sources, input modalities, and Machine Learning (ML) tasks used in this study. *SAR* refers to synthetic aperture radar data, and *InSAR* to interferometric synthetic aperture radar data.

Land use land cover scene classification

BigEarthNet[94] is a multi-class, multi-label dataset tailored for Land Use Land Cover (LULC) scene classification, consisting of Sentinel-2 optical image patches spanning 10 European countries. Each image patch in the dataset is annotated with one or more LULC class labels derived from the Corine Land Cover (CLC) database[95] (Fig. 2a). Despite its utility, BigEarthNet is subject to NS1, which stems from the methodologies used to create the CLC

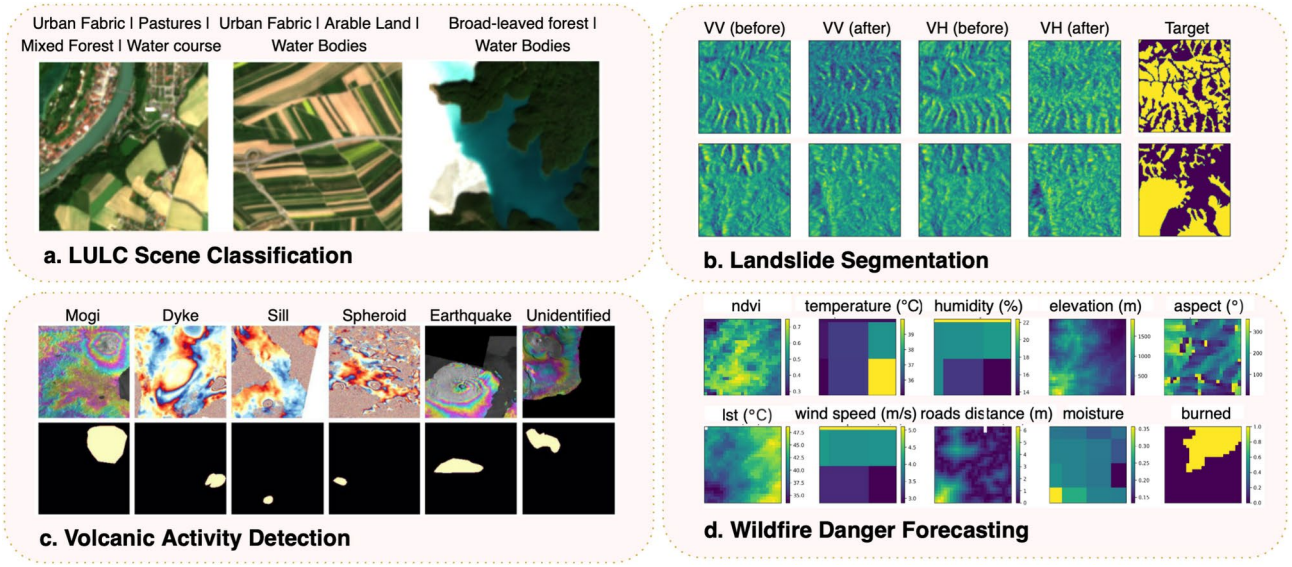


Fig. 2. Examples of data samples from the datasets used in this study, highlighting sources of label noise. (**a**) BigEarthNet: The discrepancies in labeling strategies of the Corine Land Cover database introduce label noise. (**b**) Landslides dataset: Misalignments between in-situ annotated masks and Earth observation images contribute to labeling inconsistencies. (**c**) Hephaestus: Atmospheric contributions and coherence variations challenge the annotation, leading to labels with heteroscedastic noise. (**d**) Wildfires dataset: Label noise stems from the stochastic nature of wildfire occurrence, where similar environmental conditions do not always lead to the same target class.

database. The bottom-up approach employed in CLC implementation involves the integration of national databases at the European level. Thus, the variations in annotation methodologies across different countries pose challenges to annotation consistency. Particularly, some of the countries employ semi-automatic or automatic techniques blending national in-situ data with satellite image processing, while others adhere to standard visual interpretation. These challenges are verified by empirical evaluations comparing CLC against field-based LULC mappings, indicating that the dataset's reliability falls below 90%[95]. Furthermore, these evaluations reveal the heteroscedasticity in label noise, demonstrating that certain classes, such as sparse vegetation, exhibit lower reliability compared to more consistent classes like rivers and lakes.

Landslide segmentation

The dataset used for landslide segmentation[20] comprises SAR input data paired with expert-generated landslide masks for earthquake-triggered landslides in the Hokkaido region of Japan. Figure 2b illustrates examples of such pairs. The dual polarisation SAR intensity images (VV and VH) are used to predict the landslide masks. In this dataset, the annotations for landslide masks were meticulously crafted by domain experts during fieldwork. However, precisely overlapping the SAR images with in-situ measurements is challenging, leading to misalignment between the annotated masked landslides and their actual spatial positions within the feature space. Thus, pixels that should correspond to landslides may not, and vice versa, particularly along the boundaries of the landslides. These misalignments compromise the accuracy of input-label pairs, making this dataset susceptible to NS4. Notably, the degree of inconsistency varies between pixels and samples, meaning that label noise is highly dependent on the specific image region and landslide being considered.

Volcanic activity detection

The Hephaestus dataset[21] serves as a comprehensive global InSAR repository tailored for volcanic activity monitoring. It contains a diverse set of labels, including ground deformation caused by volcanic activity, a mask depicting the location of the detected deformation, the deformation type (e.g. Sill, Dyke, Mogi) and the presence of atmospheric contributions in the InSAR imagery. Some representative samples, the ground deformation type, and the respective ground truth mask can be seen in Fig. 2c. A team of domain experts methodically executed annotation efforts for the Hephaestus dataset. They examined potential fringe patterns and, when necessary, used external sources such as Digital Elevation Models (DEMs). Nevertheless, despite the involved expertise, labeling this dataset poses challenges, primarily due to NS2. Atmospheric contributions create patterns that mimic fringes caused by actual ground deformation, complicating accurate annotation. Moreover, regions with high incoherence present significant challenges in detecting ground deformations, particularly during the early stages of volcanic unrest. The noise variability across samples induces heteroscedasticity, making some samples inherently more challenging to annotate due to variations in fringe patterns and incoherence levels.

Wildfire danger forecasting
The dataset used for the wildfire danger forecasting task[18] is derived from a spatio-temporal datacube with a daily temporal resolution and $1\text{km} \times 1\text{km}$ spatial resolution, centered around Greece. The dataset integrates variables from multiple sources, including satellite-derived vegetation status, meteorological observations, ground geomorphology data, and indicators of human activity (Fig. 2d). These variables are used to predict the likelihood that a given sample will be affected by a wildfire the subsequent day. To formulate this as a supervised ML task, samples are extracted from the spatio-temporal dataset and categorized into two classes: high and low danger. A key assumption in the sampling methodology of the referenced study is that wildfire danger increases when an actual fire incident occurs (positives) and remains low when no fire has occurred (negatives). However, wildfire occurrence is inherently stochastic, and the absence of a wildfire event does not necessarily indicate low wildfire danger[93]. To mitigate this stochasticity, negative samples are drawn from days without fire incidents across the entire geographical domain. Despite these efforts, label noise from NS3 remains a significant concern. This noise is particularly pronounced in the negative class, where samples with high wildfire risk may be incorrectly labeled as "low danger" simply because no fire occurred. Heteroscedasticity arises due to the varying noise levels across samples. For instance, negative samples collected during high-heat summer periods, when fire danger likely exists despite no recorded fire, exhibit higher noise levels compared to those collected during winter, when conditions are generally unfavorable for fire ignition.

Experiments

This section presents the experimental setups for each task and describes the evaluation methods used to assess both predictive performance and uncertainty estimation.

Experimental setup
For all four tasks, we follow the experimental setups outlined in the respective referenced papers, using the best-performing model architectures and hyperparameters for the DL models. For the Hephaestus dataset, where no established experimental protocols have been published, we design a dedicated experimental setup and evaluate multiple architectures to assess the performance of the different approaches.

For the *LULC scene classification* task, the dataset consists of 120×120 patches, with all bands of Sentinel-2 used as input. The task is framed as multi-class, multi-label image classification, where each class is predicted independently using the binary cross-entropy loss. We use the pre-defined train, validation, and test splits from BigEarthNet and employ the ResNet-50[96] model architecture.

For the *landslide segmentation* task, the dataset consists of 128×128 patches derived from dual polarisation SAR intensity images captured before and after landslide events. We use several pre-event SAR images and a single post-event image. The pre-event time series are mean-aggregated, and the VV and VH bands from both pre-and post-event images are combined into four input channels serving as input to the segmentation model. Differing from the original work, we introduce a three-way split of the data—training (70%), validation (20%), and test (10%)—to enable the tuning of the temperature parameter of the probabilistic framework. We utilize the U-Net++[97] architecture with a ResNet-50 encoder and train the model using cross-entropy loss.

For the *volcanic activity detection* task, we utilize single InSAR frames, resampled to a uniform resolution of 1024×1024. The task is formulated as a binary classification problem, where the presence or absence of ground deformation is predicted using the cross-entropy loss. A spatial split is used for dataset partitioning, selecting 12 unique locations to serve as the test set, which contains 547 samples with ground deformation and 6990 samples without deformation. The test set distribution is designed to reflect real-world conditions, simulating an operational early warning system. We employ the ResNet-18, ResNet-50, and DenseNet121[98] model architectures in our experiments.

The *wildfire danger forecasting* task is framed as a binary time series classification problem using the cross-entropy loss, with one class indicating fire danger and the other its absence. Following[18], we use a 2:1 ratio of negative (low fire danger) to positive (high fire danger) samples. Data from 2009 to 2018 are used for training, data from 2019 for validation, and data from 2020 for testing. The model implementation is based on a Long Short-Term Memory (LSTM)[99] network, using a 10-day time series of input variables.

For all tasks, the optimal temperature τ for the probabilistic framework is selected based on validation set metrics, exploring values from 0.1 to 1 in steps of 0.1, and from 1 to 10 in steps of 1. During both training and testing, we use 1000 MC samples to estimate the model's final predictions and associated uncertainties. The performance of the probabilistic framework is compared against models trained with the standard heteroscedastic approach ($\tau = 1$) and standard deterministic DL models.

Evaluation methods
In all the experiments, F_1 score and the Area Under the Precision-Recall Curve (AUPRC) are used as performance evaluation metrics. Beyond these standard metrics, a key component of our study is to assess the reliability of the models' uncertainty estimates. In line with the principle that predictions with low uncertainty should be accurate and inaccurate predictions should exhibit high uncertainty[100], we employ two methods to assess uncertainty reliability: the Discard Test[101] and Uncertainty Density Plots[102].

The *Discard test* evaluates the quality of uncertainty estimates by iteratively removing batches of the most uncertain predictions from the test set and measuring the model's error on the remaining samples. This process is repeated iteratively until all samples have been discarded, producing a curve that illustrates how the model error changes with each iteration. A model that produces reliable uncertainty estimates should exhibit decreasing error as more uncertain predictions are discarded, indicating that more uncertain samples correspond to less accurate predictions. In this study, we use the loss as an indicator of error and consider 10 discard fractions. The results of the Discard Test are visualized using a line plot that displays the discard fraction alongside the model's

error, along with two key metrics: Monotonicity Fraction (MF) and Discard Improvement (DI). MF measures how often model performance improves as more uncertain samples are discarded. It is computed as:

$$MF = \frac{1}{N_f - 1} \sum_{i=1}^{N_f - 1} I(\epsilon_i \geq \epsilon_{i+1}),$$

where I is the indicator function, ϵ_i is the model error (here the loss) at discard fraction i, and N_f denotes the total number of considered discard fractions. An MF value of 1 indicates perfect monotonicity. DI quantifies the average reduction in model error as the discard fraction increases and is given by:

$$DI = \frac{1}{N_f - 1} \sum_{i=1}^{N_f - 1} I(\epsilon_i - \epsilon_{i+1}).$$

The *Uncertainty density plots* illustrate the distribution of uncertainty scores for test set samples, distinguishing between correctly and incorrectly classified instances. The median uncertainty for each group is also reported. In a reliable model, misclassified samples are expected to exhibit higher uncertainty, whereas correctly classified samples should show lower uncertainty, leading to a clear separation between the distributions.

Results and discussion

Evaluation performance of the ML models

The evaluation performance metrics for all models are presented in Table 2. For all applications, the F_1 score and the AUPRC are reported separately. For the binary classification tasks F_1 score is reported for the positive class, while for the multi-class datasets, the micro F_1 score is used.

Despite the differences in input modalities, ML setups, and sources of label noise, the probabilistic framework consistently improves performance across most tasks. Notably, it achieves the highest scores in LULC scene classification and wildfire danger forecasting and outperforms deterministic models in volcanic activity detection when using ResNet-50 and DenseNet121. It fails to surpass deterministic models in AUPRC for landslide segmentation, despite achieving a significantly higher F_1 score. Additionally, it underperforms in both metrics for volcanic activity detection when using ResNet-18. Furthermore, except for the F_1 score in wildfire danger forecasting, the probabilistic model outperforms the standard heteroscedastic model. In some cases, the optimal τ is found to be 1, indicating that the probabilistic and heteroscedastic models coincide. In general, the best τ varies across datasets and model architectures, emphasizing the need for its careful tuning for improved performance.

The decrease in AUPRC for landslide segmentation can be attributed to the increased variability introduced by the probabilistic model—which relies on multiple MC samples per prediction. While this approach improves robustness to label noise, it can also reduce confidence in predictions, particularly in ambiguous regions such as landslide boundaries, where label noise is concentrated. This can result in fewer true positives identified with high confidence, leading to a lower AUPRC in this specific application. Since AUPRC evaluates performance across multiple thresholds, it is more sensitive to these confidence variations. In contrast, the F_1 score, computed using a fixed threshold, remains unaffected.

The most significant performance improvement is observed in the task of volcanic activity detection. This enhancement can be attributed to the task's inherent challenges, particularly the high similarity between positive and negative atmospheric InSAR signals, which introduces substantial uncertainty during labeling. In this context, the probabilistic approach has effectively captured this labeling uncertainty, resulting in a marked improvement in the performance metrics of the larger models i.e. ResNet-50 and DenseNet121.

Task	Model	Deterministic DL		Standard heteroscedastic ($\tau = 1$)		Probabilistic ML ($\tau = \tau^*$)	
		F_1 Score	AUPRC	F_1 Score	AUPRC	F_1 Score	AUPRC
LULC Scene Classification	ResNet-50 ($\tau^* = 1$)	0.748	0.816	**0.754**	**0.829**	**0.754**	**0.829**
Landslide Segmentation	U-Net ($\tau^* = 2$)	0.545	**0.660**	0.476	0.648	**0.593**	0.640
Wildfire Danger Forecasting	LSTM ($\tau^* = 0.2$)	0.767	0.872	**0.777**	0.879	0.776	**0.883**
Volcanic Activity Detection	ResNet-18 ($\tau^* = 5$)	**0.402**	**0.357**	0.315	0.261	0.387	0.304
	ResNet-50 ($\tau^* = 0.9$)	0.436	0.437	0.428	0.435	**0.518**	**0.518**
	DenseNet121 ($\tau^* = 2$)	0.339	0.359	**0.495**	**0.468**	**0.495**	**0.468**

Table 2. Comparison of model performance using probabilistic ML[24], standard heteroscedastic model[23], and deterministic DL across multiple tasks and model architectures. The optimal temperature scaling parameter τ^* is shown per task/architecture. Best values per task, metric, and architecture are highlighted in bold.

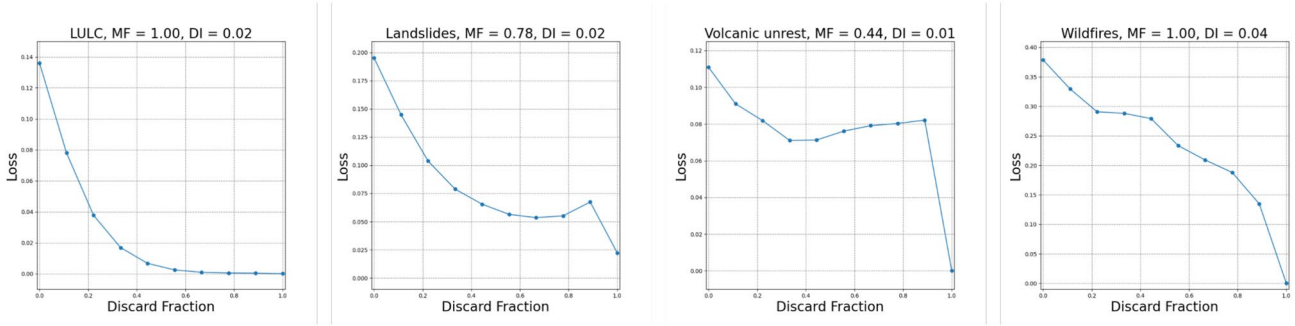


Fig. 3. Discard test plots across all tasks. A reliable model should exhibit a decreasing error trend as the discard fraction increases, indicating that the most uncertain samples correspond to higher loss values. The Monotonicity Fraction (MF) measures the frequency with which the error decreases upon discarding uncertain samples, while the Discard Improvement (DI) quantifies the average reduction in model error as the discard fraction increases.

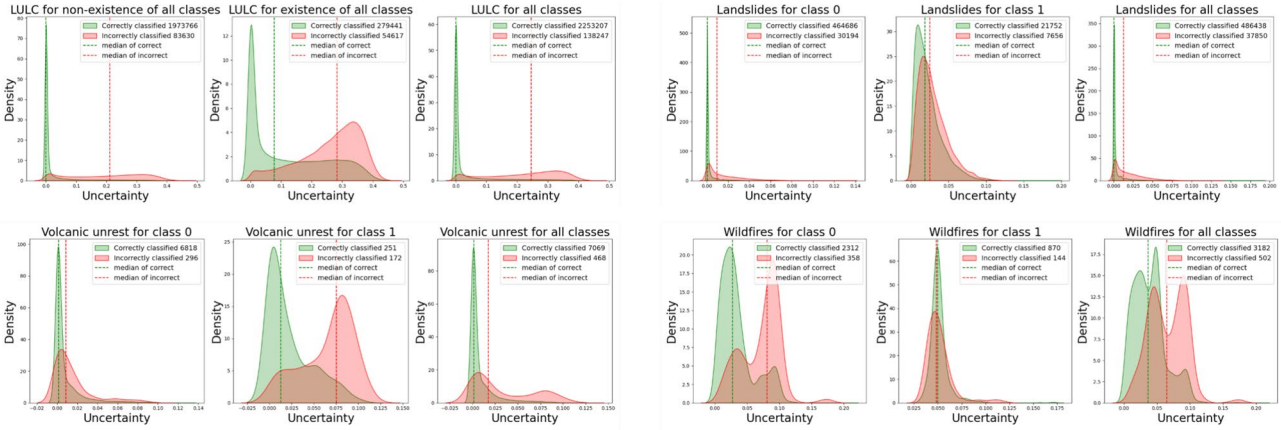


Fig. 4. Uncertainty density plots across all tasks, presented for all classes combined and separately for the positive and negative classes. Vertical dashed lines indicate the median uncertainty for each group. Reliable uncertainty estimates are characterized by distinct, well-separated distributions with minimal overlap.

Uncertainty evaluation

Beyond optimizing predictive accuracy, this study primarily aims to evaluate the reliability of the uncertainty estimates produced by the probabilistic model. To achieve this, we assess the quality of these estimates using Discard Test plots (Fig. 3) and Uncertainty Density plots (Fig. 4).

Figure 3 illustrates the Discard Test plots for all tasks, where the most uncertain samples are progressively removed, and the model's loss is calculated on the remaining samples. Across all tasks, the discard test exhibits the optimal behavior, with model loss consistently decreasing as more uncertain samples are removed. This indicates a clear alignment between high uncertainty and high-loss samples, highlighting the reliability of uncertainty estimates across the diverse applications in this study. In volcanic activity detection, a slight increase in loss occurs after discarding more than half of the most uncertain samples (discard fraction ≥ 0.5), before decreasing again in the last values. This increase appears more like a plateau rather than a significant rise in loss. This behavior is likely due to the dataset's high imbalance, where most samples correspond to non-deformation cases that the model classifies with high confidence (low loss) and low uncertainty (see Fig. 4, third plot). Despite variations in loss functions and scales across tasks, DI remains positive, indicating a consistent overall reduction in loss. Moreover, MF further supports these findings, reaching a perfect score (1) for LULC scene classification and wildfire danger forecasting while achieving 0.78 for landslide segmentation. In volcanic activity detection, the observed increase in loss impacts MF, resulting in a lower score of 0.44.

Figure 4 presents the uncertainty density plots for all tasks, illustrating the distribution of the predicted uncertainties for correctly and incorrectly classified samples. For each application, three plots are provided. In binary classification tasks, the first plot represents the uncertainty densities for negative labels, the second for positive labels, and the third for all. For the multi-label LULC task, the first plot shows the uncertainty densities for all labels assigned as 0 (absent in the sample), the second for all labels assigned as 1 (present in the sample), and the third combines them all. This separation between negative and positive labels helps to reveal any class-specific variations in uncertainty reliability.

As shown in the "all classes" plots (third column), the probabilistic model consistently assigns higher uncertainty to misclassified instances, in line with the principle that predictions with lower uncertainty should be accurate, while inaccurate predictions should exhibit higher uncertainty. While this trend holds for all tasks, a more detailed analysis of the plots is conducted to extract task-specific insights.

For the LULC task, the observed trend persists across all classes, indicating high reliability in this task. In landslide segmentation, correctly classified samples from class 0 (non-landslide areas) exhibit near-zero uncertainty, suggesting that these image regions are minimally affected by label noise. In contrast, misclassified samples show higher uncertainty, a trend also reflected in the median values. Class 1 (landslide areas) displays wider uncertainty distributions for both correctly and incorrectly classified samples. This indicates that this class is more prone to noise and is more challenging for the model, as also illustrated in Fig. 5b. Despite these challenges, misclassified samples still exhibit higher uncertainty, highlighting the model's ability to quantify uncertainty reliably in this application.

In volcanic activity detection, a clear separation in uncertainty is observed for class 1 (ground deformation). For class 0 (no ground deformation), uncertainty estimates are generally lower for both correctly and incorrectly classified samples, suggesting that this class exhibits less noise. However, as indicated in the figure legend, only a small proportion of class 0 samples is misclassified, implying that this class is relatively easy for the model to classify. Despite smaller differences in uncertainty distributions for class 0, the mode and median values for misclassified samples remain higher, showing the model's ability to distinguish uncertainty even in this class.

In wildfire danger forecasting, the uncertainty distributions are more complex. For class 0 (low danger), while the separation between correct and incorrect classifications is clear, a bimodal distribution emerges for both groups. One mode exhibits near-zero uncertainty, while the other lies in higher uncertainty values. This pattern reflects the seasonal composition of the dataset, where negative samples from different periods (e.g. summer vs winter) exhibit varying noise levels, which the model appears to have learned effectively. For class 1 (high danger), overall uncertainty remains low, with the model struggling to distinguish between correct and incorrect classifications. This behavior aligns with the expectation that positive samples (indicating actual fire danger) exhibit lower noise levels. However, the persistence of low uncertainty even in the misclassified samples, suggests that the inherent characteristics of the data have limited the model's ability to express high uncertainty in this application, even for incorrect predictions.

Qualitative assessment of aleatoric uncertainty

Figure 5 presents representative samples from all datasets, demonstrating how the predicted uncertainty estimates can provide additional insights for each task.

Figure 5-a presents examples from the BigEarthNet dataset with the highest predicted uncertainty. The three images on the right correspond to the samples with the highest average uncertainty across all classes. These samples exhibit multiple visually ambiguous classes, showing that the model is less confident when providing predictions for samples with complex or overlapping land cover types. The three images on the left depict samples where the model predicted with high uncertainty the presence of inland wetlands, the class with the highest individual uncertainty among all. This class is absent from the ground-truth labels in the first two images but present in the third. This difficulty can be attributed to the inherent challenges in distinguishing inland waters in EO images, where spectral overlap with vegetation, forest, or bare soil leads to ambiguous classification, complicating even visual identification[103].

Figure 5-b presents the predicted aleatoric uncertainty alongside segmentation maps for two landslide scenarios. The model consistently assigns higher uncertainty at the edges of the landslides, where annotation errors are more likely to occur.

Figure 5-c presents eight samples from the Hephaestus dataset, categorized by ground deformation presence and predicted uncertainty levels. In low-uncertainty samples, ground deformation is easily distinguishable, with InSAR data exhibiting distinct fringes, while no deformation regions appear smooth with minor perturbations, likely due to atmospheric effects. High-uncertainty samples, however, are more challenging. In the first no deformation sample, the source of uncertainty is unclear, whereas in the second, a prominent fringe likely confused the model, leading to lower-confidence predictions. For ground deformation cases, in the first sample, the deformation pattern is subtle, justifying the high-uncertainty prediction. In the second, the InSAR is poorly overlaid on a DEM, distorting the output and confusing the model. Interestingly, since Hephaestus maps InSAR data to pixel space, visualization choices, such as colormap selection and DEM overlays, can influence predictions by altering the physical meaning of the data.

Figure 5-d presents a wildfire prediction map alongside its predicted aleatoric uncertainty. Uncertainty is high in regions with intermediate wildfire danger probabilities, reflecting the model's ambiguity in these cases, while it remains low for extreme probabilities near 0 and 1. Although a relationship between softmax outputs and aleatoric uncertainty may be observed in this task, the two measures are inherently complementary: softmax outputs reflect the model's most likely prediction, whereas aleatoric uncertainty quantifies the reliability of that prediction in the presence of input-dependent noise.

No sign of epistemic uncertainty

In this study, we employed a probabilistic ML model specifically designed to capture aleatoric uncertainty. To validate the absence of epistemic uncertainty in the predicted uncertainty estimates, we evaluate their sensitivity in out-of-distribution (OOD) conditions for the wildfire danger forecasting task.

The model for this task was originally trained on data from Greece up to 2018, with 2019 used for validation and 2020 as the in-distribution test set. To investigate the uncertainty's behavior under distributional shifts, we incorporate two additional test sets. The first consists of Greek data from 2021, a year marked by an extreme fire season in Greece, characterized by prolonged heatwaves and unusually low humidity, which was the worst of

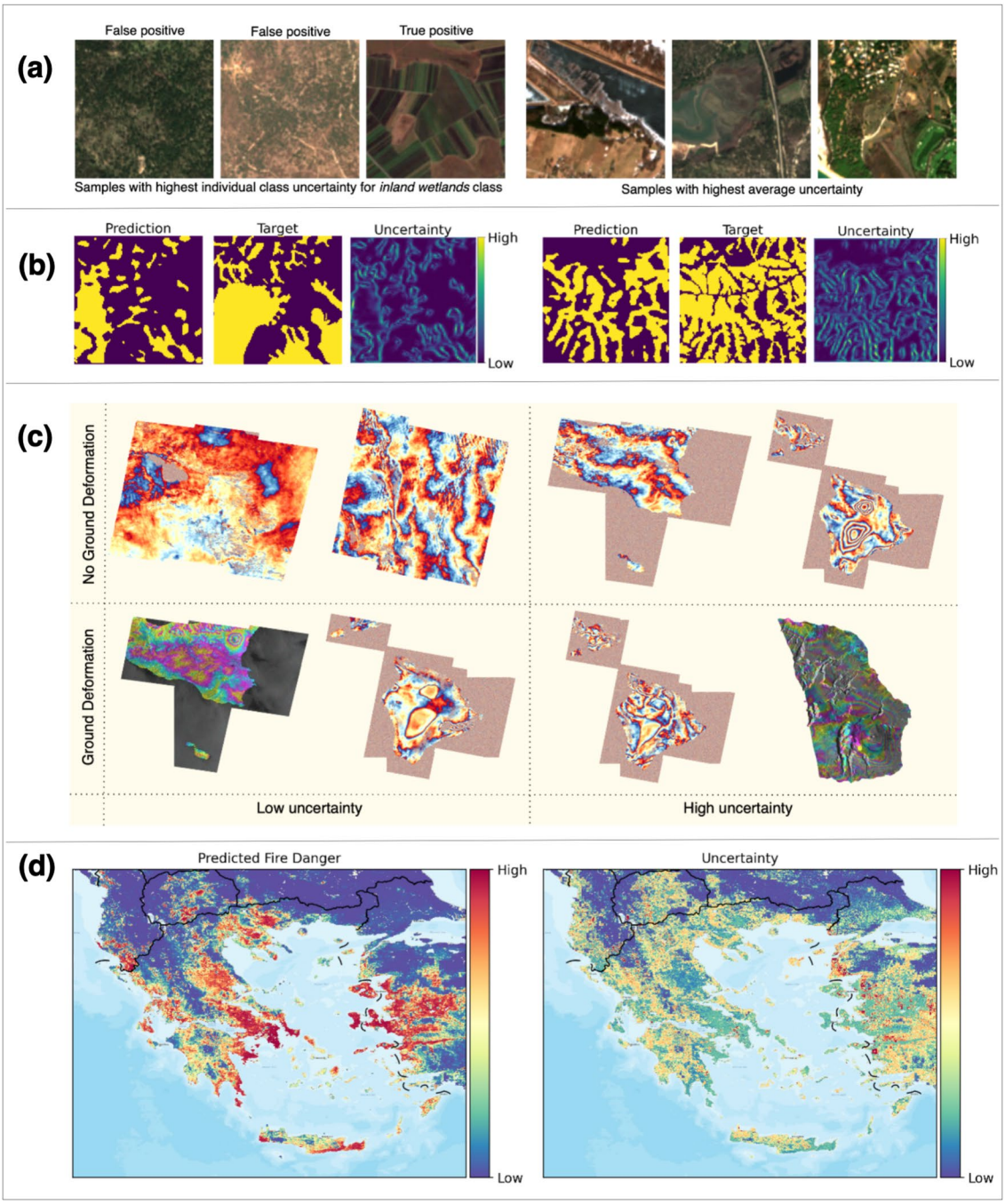


Fig. 5. Samples with low and high uncertainty for all tasks. (**a**) Example of uncertain samples from BigEarthNet, licensed under CDLA-P 1.0. The left panel shows samples with the highest uncertainty from the *inland wetlands* class, which exhibits the greatest individual uncertainty among all classes. The right panel displays samples with the highest average uncertainty across all classes; (**b**) ground truth and predicted segmentation maps for two landslide events, along with the predicted aleatoric uncertainty. The uncertainty is higher at the boundaries of the landslides, which represent the most challenging regions during manual annotation; (**c**) samples from the Hephaestus dataset, categorized based on the presence of ground deformation and corresponding predicted uncertainty levels; (**d**) a wildfire danger prediction map accompanied by the predicted aleatoric uncertainty. These uncertainty maps can provide valuable insights for decision-makers, highlighting the confidence level of the machine learning models in their forecast. The maps were generated in Python v3.8.20 using matplotlib v3.7.5 (https://matplotlib.org) and Basemap v1.3.6 (https://matplotlib.org/basemap/).

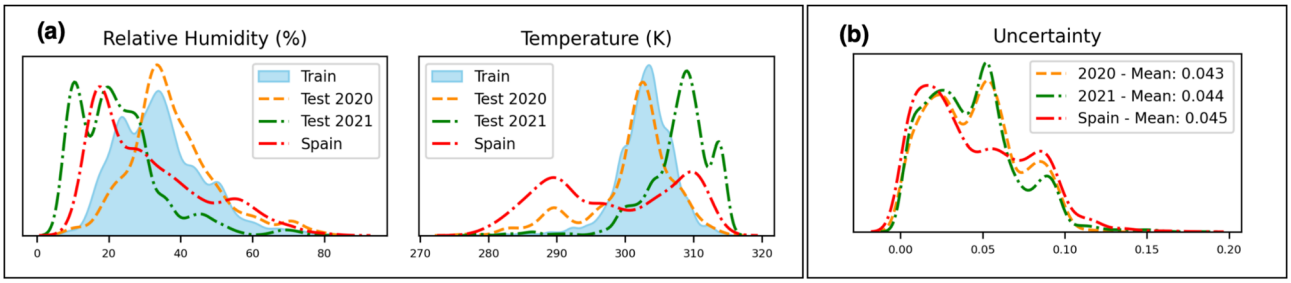


Fig. 6. Comparison of environmental variables and predicted uncertainties across multiple test sets for the wildfire danger forecasting task. (**a**) Relative humidity and temperature distributions for the burned pixels of Greek training data, 2020 in-distribution test set, 2021 Greek out-of-distribution data, and Spanish out-of-distribution dataset. (**b**) Corresponding predicted uncertainty distributions across all test sets.

the past 13 years[104]. The second test set contains data from Spain, a region with distinct geographic and climatic characteristics compared to Greece, extracted from the Mesogeos dataset[105]. Figure 6a presents the distributions for temperature and relative humidity, two of the most critical fire drivers, across the Greek training data, the 2020 in-distribution test, the 2021 Greek OOD data, and the Spanish OOD dataset. The figure highlights the distributional shifts observed in both the 2021 Greek and Spanish datasets relative to the training and 2020 test data.

Since the data generation process is consistent across all datasets, the label noise, and therefore the inherent aleatoric uncertainty, is expected to be similar across them. Consequently, if the model were sensitive to epistemic uncertainty, an increase in predicted uncertainty should be observed on OOD datasets, specifically attributed to this additional uncertainty source. However, as shown in Fig. 6b, the uncertainty distributions for the 2020 in-distribution and 2021 OOD Greek datasets are nearly indistinguishable, while the Spanish dataset exhibits only small deviations. Moreover, the mean uncertainty values are almost identical across all sets, indicating that the model does not capture epistemic uncertainty.

Role of uncertainty in decision-making

Trustworthiness is a fundamental requirement for the successful deployment of DL models in operational environments. In this study, we leveraged uncertainty-aware models to address this critical need, demonstrating their reliability across various disaster management scenarios. While model predictions alone are useful for guiding decision-making, incorporating uncertainty provides valuable complementary insights into their reliability. This additional information can be particularly important in contexts where decisions must be made rapidly and resources are limited. For example, interventions can be prioritized in areas exhibiting both high predicted danger outputs and low uncertainty, while high uncertainty regions may warrant further data collection, increased caution, or consultation with domain experts. In volcanic activity monitoring, uncertainty-informed predictions can enhance eruption preparedness by guiding urgent evacuations near volcanoes with high predicted eruption risk and low uncertainty, while enabling a cautious management in cases with lower predicted risk but high uncertainty. Similarly, in wildfire danger forecasting, uncertainty-informed predictions allow for more strategic resource allocation by directing firefighting efforts to areas with higher predicted fire danger and lower uncertainty, which can optimize operations and reduce costs. For landslide segmentation, uncertainty estimation can guide emergency responders in prioritizing search and rescue operations by focusing on areas with high predicted landslide probability and lower prediction uncertainty. Overall, by integrating uncertainty estimation, DL EO models transition from "black-box" systems to more transparent and trustworthy tools, ultimately supporting more robust and reliable decision-making in real-world applications.

Limitations & opportunities

This study focused on addressing aleatoric uncertainty, enhancing reliability in EO tasks. However, epistemic uncertainty, another critical component for trustworthy EO applications, was not modeled and remains a valuable direction for future work. Moreover, our findings revealed that the impact of probabilistic modeling on predictive performance varied across tasks, models, and metrics. For instance, in the volcanic activity detection task, deeper architectures exhibited significant gains from the probabilistic ML models, while shallower models did not. Conversely, in wildfire danger forecasting, where a relatively simple LSTM model was used, probabilistic modeling improved performance. In landslide segmentation, we observed improvements in the F1 Score but not in AUPRC, highlighting how the choice of evaluation metric can affect the assessment of model improvements. These results emphasize the importance of task-specific understanding to identify the conditions under which the probabilistic ML model is effective and to ensure meaningful interpretation of both uncertainty estimates and model predictions. Looking forward, the integration of reliable uncertainty quantification opens avenues for advancing ML in other EO domains. These include domain adaptation[106], active learning[107], or semi-supervised[108] learning applications. Given the inherent variability in EO data, with variations in scale, resolution, and environmental conditions[109], incorporating uncertainty-driven sampling and adaptation strategies holds significant potential for these domains.

Conclusion

In this work, we demonstrated the potential of uncertainty-aware ML in addressing input-dependent label noise in EO. Through four high-stakes EO applications, particularly in disaster management, we highlighted how these approaches enhance the reliability of DL models in EO, an essential requirement for deploying ML solutions in real-world scenarios. Our proposed pipeline, ranging from model performance assessment to the evaluation of uncertainty estimates and their interpretation in decision-making processes, holds significant potential for broader applications across various EO domains that require uncertainty quantification. A key limitation of this work is its exclusive focus on aleatoric uncertainty estimation. Future work will extend the proposed framework to include both aleatoric and epistemic uncertainty, further advancing the development of more robust, interpretable, and reliable ML-driven solutions in EO.

Data availability

All data used in this study are publicly available. Detailed information on how to access and download the datasets can be found in the original publications referenced for each of the tasks included in this study. Any specific requests for data and materials should be addressed to S.K.

Received: 4 July 2025; Accepted: 10 September 2025
Published online: 14 October 2025

References

1. Soille, P. et al. A versatile data-intensive computing platform for information retrieval from big geospatial data. *Future Gen. Comput. Syst.* **81**, 30–40. https://doi.org/10.1016/j.future.2017.11.007 (2018).
2. Reichstein, M. et al. Deep learning and process understanding for data-driven Earth system science. *Nature* **566**, 195–204. https://doi.org/10.1038/s41586-019-0912-1 (2019) (publisher: Nature Publishing Group).
3. Yang, J. et al. The role of satellite remote sensing in climate change studies. *Nat. Clim. Change* **3**, 875–883 (2013).
4. Nakalembe, C. Urgent and critical need for sub-Saharan African countries to invest in earth observation-based agricultural early warning and monitoring systems. *Environ. Res. Lett.* **15**, 121002 (2020).
5. Wu, Z. et al. Deep learning enables satellite-based monitoring of large populations of terrestrial mammals across heterogeneous landscape. *Nat. Commun.* **14**, 3072. https://doi.org/10.1038/s41467-023-38901-y (2023) (publisher: Nature Publishing Group).
6. Camps-Valls, G., Tuia, D., Zhu, X. & Reichstein, M. *Deep Learn. Earth Sci.* https://doi.org/10.1002/9781119646181 (2021).
7. Zhu, X. X. et al. Deep learning in remote sensing: A comprehensive review and list of resources. *IEEE Geosci. Remote Sens. Mag.* **5**, 8–36. https://doi.org/10.1109/MGRS.2017.2762307 (2017).
8. Ma, L. et al. Deep learning in remote sensing applications: A meta-analysis and review. *ISPRS J. Photogramm. Remote Sens.* **152**, 166–177. https://doi.org/10.1016/j.isprsjprs.2019.04.015 (2019).
9. Camps-Valls, G., Tuia, D., Zhu, X. X. & Reichstein, M. *Deep Learning for the Earth Sciences: A Comprehensive Approach to Remote Sensing, Climate Science, and Geosciences* (Wiley, 2021).
10. Zhang, L., Zhang, L. & Du, B. Deep learning for remote sensing data: A technical tutorial on the state of the art. *IEEE Geosci. Remote Sens. Mag.* **4**, 22–40. https://doi.org/10.1109/MGRS.2016.2540798 (2016).
11. Aleissaee, A. A. et al. Transformers in remote sensing: A survey. *Remote Sens.* **15**. https://doi.org/10.3390/rs15071860 (2023).
12. LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* **521**, 436–444. https://doi.org/10.1038/nature14539 (2015) (number: 7553 publisher: Nature Publishing Group).
13. Sumbul, G. & Demir, B. Generative reasoning integrated label noise robust deep image representation learning. *IEEE Trans. Image Process.* **32**, 4529–4542 (2022).
14. Elmes, A. et al. Accounting for training data error in machine learning applied to Earth observations. *Remote Sens.* **12**, 1034. https://doi.org/10.3390/rs12061034 (2020) (number: 6 publisher: Multidisciplinary Digital Publishing Institute).
15. Jiang, L., Zhou, Z., Leung, T., Li, L.-J. & Fei-Fei, L. MentorNet: Learning data-driven curriculum for very deep neural networks on corrupted labels. In *Proceedings of the 35th International Conference on Machine Learning*. 2304–2313 (PMLR, 2018) (ISSN: 2640-3498).
16. Gawlikowski, J. et al. A survey of uncertainty in deep neural networks. *Artif. Intell. Rev.* **56**, 1513–1589. https://doi.org/10.1007/s10462-023-10562-9 (2023).
17. Tuia, D. et al. Toward a collective agenda on AI for Earth science data analysis. *IEEE Geosci. Remote Sens. Mag.* **9**, 88–104. https://doi.org/10.1109/MGRS.2020.3043504 (2021).
18. Kondylatos, S. et al. Wildfire danger prediction and understanding with deep learning. *Geophys. Res. Lett.* **49**, e2022GL099368. https://doi.org/10.1029/2022GL099368 (2022). _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1029/2022GL099368.
19. Bountos, N. I. et al. Kuro siwo: 33 billion m^2 under the water. A global multi-temporal satellite dataset for rapid flood mapping. In *Advances in Neural Information Processing Systems* (Globerson, A. et al. eds.). Vol. 37. 38105–38121 (Curran Associates, Inc., 2024).
20. Boehm, V. et al. Deep Learning for Rapid Landslide Detection Using Synthetic Aperture Radar (SAR) Datacubes (2022). arXiv:2211.02869 [cs, eess].
21. Bountos, N. I. et al. Hephaestus: A large scale multitask dataset towards InSAR understanding. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 1452–1461. https://doi.org/10.1109/CVPRW56347.2022.00151 (IEEE, 2022).
22. Gal, Y. *Uncertainty in Deep Learning* (PhD thesis, University of Cambridge, 2016).
23. Kendall, A. & Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS'17. 5580–5590 (Curran Associates Inc., 2017).
24. Collier, M., Mustafa, B., Kokiopoulou, E., Jenatton, R. & Berent, J. A Simple Probabilistic Method for Deep Classification Under Input-Dependent Label Noise (2020). arXiv:2003.06778 [cs, stat].
25. Cordeiro, F. R. & Carneiro, G. A survey on deep learning with noisy labels: How to train your model when you cannot trust on the annotations? In *2020 33rd SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)*. 9–16. https://doi.org/10.1109/SIBGRAPI51738.2020.00010 (IEEE, 2020).
26. Tanno, R., Saeedi, A., Sankaranarayanan, S., Alexander, D. C. & Silberman, N. Learning from noisy labels by regularized estimation of annotator confusion. In *CVPR*. 11244–11253 (2019).
27. Menon, A. K., Rawat, A. S., Kumar, S. & Reddi, S. Can gradient clipping mitigate label noise? In *International Conference on Learning Representations (ICLR)* (2020).
28. Wei, H., Tao, L., Xie, R. & An, B. Open-set label noise can improve robustness against inherent label noise. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*, NIPS '21 (Curran Associates Inc., 2021).

29. Hendrycks, D., Mazeika, M., Wilson, D. & Gimpel, K. Using trusted data to train deep networks on labels corrupted by severe noise. In *Advances in Neural Information Processing Systems* (Bengio, S. et al. eds.) . Vol. 31 (Curran Associates, Inc., 2018).
30. Zhang, Y. & Sugiyama, M. Approximating Instance-Dependent Noise via Instance-Confidence Embedding. https://doi.org/10.48550/arXiv.2103.13569 (2021). arXiv:2103.13569 [cs, stat].
31. Yang, S. et al. Estimating instance-dependent Bayes-label transition matrix using a deep neural network. In *ICML* (2022).
32. Wang, R., Liu, T. & Tao, D. Multiclass Learning with Partially Corrupted Labels. *IEEE Trans. Neural Netw. Learn. Syst.* **29**, 2568–2580. https://doi.org/10.1109/TNNLS.2017.2699783 (2018) (conference name: IEEE Transactions on Neural Networks and Learning Systems).
33. Chang, H.-S., Learned-Miller, E. & McCallum, A. Active bias: Training more accurate neural networks by emphasizing high variance samples. In *Advances in Neural Information Processing Systems*. Vol. 30 (Curran Associates, Inc., 2017).
34. Zhang, H., Xing, X. & Liu, L. DualGraph: A graph-based method for reasoning about label noise. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 9649–9658. https://doi.org/10.1109/CVPR46437.2021.00953 (IEEE, 2021).
35. Song, H., Kim, M., Park, D., Shin, Y. & Lee, J.-G. Learning from noisy labels with deep neural networks: A survey. *IEEE Trans. Neural Netw. Learn. Syst.* **34**, 8135–8153. https://doi.org/10.1109/tnnls.2022.3152527 (2023).
36. Jiang, L., Huang, D. , Liu, M. & Yang, W. Deep learning on controlled noisy labels. In *ICML, Beyond Synthetic Noise* (2020).
37. Zhang, H., Cissé, M., Dauphin, Y. N. & Lopez-Paz, D. mixup: Beyond empirical risk minimization. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings* (OpenReview.net, 2018).
38. Han, B. et al. Co-teaching: robust training of deep neural networks with extremely noisy labels. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, NIPS'18. 8536–8546 (Curran Associates Inc., 2018).
39. Detlefsen, N. S., Jørgensen, M. & Hauberg, S. *Reliable Training and Estimation of Variance Networks* (Curran Associates Inc., 2019).
40. Zhang, J., Kailkhura, B. & Han, T. Y.-J. Mix-n-match: Ensemble and compositional methods for uncertainty calibration in deep learning. In *Proceedings of the 37th International Conference on Machine Learning*, ICML'20 (JMLR.org, 2020).
41. Thulasidasan, S., Chennupati, G., Bilmes, J. A., Bhattacharya, T. & Michalak, S. On mixup training: Improved calibration and predictive uncertainty for deep neural networks. In *Advances in Neural Information Processing Systems* (Wallach, H. et al. eds.). Vol. 32 (Curran Associates, Inc., 2019).
42. Angelopoulos, A. N. & Bates, S. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification, https://doi.org/10.48550/arXiv.2107.07511 (2022). ArXiv:2107.07511 [cs].
43. Malinin, A. & Gales, M. Predictive uncertainty estimation via prior networks. In *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, NIPS'18. 7047–7058 (Curran Associates Inc., 2018).
44. Jospin, L. V., Laga, H., Boussaid, F., Buntine, W. & Bennamoun, M. Hands-on Bayesian neural networks-a tutorial for deep learning users. *IEEE Comput. Intell. Mag.* **17**, 29–48. https://doi.org/10.1109/MCI.2022.3155327 (2022).
45. Blundell, C., Cornebise, J., Kavukcuoglu, K. & Wierstra, D. Weight uncertainty in neural networks. In *Proceedings of the 32nd International Conference on International Conference on Machine Learning*, ICML'15. Vol. 37. 1613–1622 (JMLR.org, 2015).
46. Lakshminarayanan, B., Pritzel, A. & Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS'17. 6405–6416 (Curran Associates Inc., 2017).
47. Ramalho, T. & Miranda, M. Density estimation in representation space to predict model uncertainty (2019). arXiv:1908.07235 [cs, stat].
48. Oberdiek, P., Rottmann, M. & Gottschalk, H. Classification uncertainty of deep neural networks based on gradient information. In *Artificial Neural Networks in Pattern Recognition: 8th IAPR TC3 Workshop, ANNPR 2018, Siena, Italy, September 19-21, 2018, Proceedings*. 113–125. https://doi.org/10.1007/978-3-319-99978-4_9 (Springer, 2018).
49. Lee, J. & AlRegib, G. Gradients as a measure of uncertainty in neural networks. In *2020 IEEE International Conference on Image Processing (ICIP)*. 2416–2420. https://doi.org/10.1109/ICIP40778.2020.9190679 (2020).
50. Sensoy, M., Kaplan, L. & Kandemir, M. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems* (Bengio, S. et al. eds.). Vol. 31 (Curran Associates, Inc., 2018).
51. Nandy, J., Hsu, W. & Lee, M. Towards maximizing the representation gap between in-domain & out-of-distribution examples. In *NeurIPS* (2020).
52. Ayhan, M. S. & Berens, P. Test-time data augmentation for estimation of heteroscedastic aleatoric uncertainty in deep neural networks. In *1st Conference on Medical Imaging with Deep Learning (MIDL)* (2018).
53. Wang, G. *et al.* Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing* **338**, 34–45. https://doi.org/10.1016/j.neucom.2019.01.103 (2019). ArXiv:1807.07356 [cs].
54. Rasmussen, C. E. & Williams, C. K. I. *Gaussian processes for machine learning. Adaptive computation and machine learning* (MIT Press, 2006). OCLC: ocm61285753.
55. Damianou, A. & Lawrence, N. Deep Gaussian processes. In *Artificial Intelligence and Statistics*. 207–215 (2013).
56. Camps-Valls, G. et al. A survey on Gaussian processes for Earth-observation data analysis: A comprehensive investigation. *IEEE Geosci. Remote Sens. Mag.* **4**, 58–78. https://doi.org/10.1109/MGRS.2015.2510084 (2016).
57. Camps-Valls, G., Sejdinovic, D., Runge, J. & Reichstein, M. A perspective on Gaussian processes for Earth observation. *Natl. Sci. Rev.* **6**, 616–618. https://doi.org/10.1093/nsr/nwz028 (2019).
58. Mateo-Sanchis, A., Munoz-Mari, J., Campos-Taberner, M., Garcia-Haro, J. & Camps-Valls, G. Gap filling of biophysical parameter time series with multi-output Gaussian processes. In *IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium*. 4039–4042. https://doi.org/10.1109/IGARSS.2018.8519254 (IEEE, 2018).
59. Svendsen, D. H., Morales-Álvarez, P., Ruescas, A. B., Molina, R. & Camps-Valls, G. Deep Gaussian processes for biogeophysical parameter retrieval and model inversion. *ISPRS J. Photogram. Remote Sens.* **166**, 68–81. https://doi.org/10.1016/j.isprsjprs.2020.04.014 (2020).
60. Morales-Alvarez, P., Perez-Suay, A., Molina, R. & Camps-Valls, G. Remote sensing image classification with large-scale gaussian processes. *IEEE Trans. Geosci. Remote Sens.* **56**, 1103–1114. https://doi.org/10.1109/TGRS.2017.2758922 (2018).
61. Gal, Y. & Ghahramani, Z. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning* (Balcan, M. F. & Weinberger, K. Q. eds.). Vol. 48. *Proceedings of Machine Learning Research*. 1050–1059 (PMLR, 2016).
62. Martínez-Ferrer, L. et al. Quantifying uncertainty in high resolution biophysical variable retrieval with machine learning. *Remote Sens. Environ.* **280**, 113199. https://doi.org/10.1016/j.rse.2022.113199 (2022).
63. Asadi, N., Scott, K. A., Komarov, A. S., Buehner, M. & Clausi, D. A. Evaluation of a neural network with uncertainty for detection of ice and water in SAR imagery. In *IEEE Transactions on Geoscience and Remote Sensing*. 1–13. https://doi.org/10.1109/TGRS.2020.2992454 (2020).
64. Kampffmeyer, M., Salberg, A. & Jenssen, R. Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks. In *2016 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 680–688. https://doi.org/10.1109/CVPRW.2016.90 (2016) (ISSN: 2160-7516).
65. Dechesne, C., Lassalle, P. & Lefèvre, S. Bayesian U-Net: Estimating uncertainty in semantic segmentation of Earth observation images. *Remote Sens.* **13**, 3836. https://doi.org/10.3390/rs13193836 (2021) (number: 19 publisher: Multidisciplinary Digital Publishing Institute).

66. He, X., Chen, Y. & Huang, L. Bayesian deep learning for hyperspectral image classification with low uncertainty. *IEEE Trans. Geosci. Remote Sens.* **61**, 1–16. https://doi.org/10.1109/TGRS.2023.3257865 (2023).
67. Joshaghani, M., Davari, A., Hatamian, F. N., Maier, A. & Riess, C. Bayesian convolutional neural networks for limited data hyperspectral remote sensing image classification. *IEEE Geosci. Remote Sens. Lett.* **20**, 1–5. https://doi.org/10.1109/LGRS.2023.3287504 (2023).
68. Ma, Y., Zhang, Z., Kang, Y. & Özdoğan, M. Corn yield prediction and uncertainty analysis based on remotely sensed variables using a Bayesian neural network approach. *Remote Sens. Environ.* **259**, 112408. https://doi.org/10.1016/j.rse.2021.112408 (2021).
69. Li, Y., Yan, S. & Gong, J. Quantifying uncertainty in soil moisture retrieval using a Bayesian neural network framework. *Comput. Electron. Agric.* **215**, 108414. https://doi.org/10.1016/j.compag.2023.108414 (2023).
70. Zhang, C. et al. SelectSeg: Uncertainty-based selective training and prediction for accurate crack segmentation under limited data and noisy annotations. *Reliab. Eng. Syst. Saf.* **259**, 110909. https://doi.org/10.1016/j.ress.2025.110909 (2025).
71. He, Q., Qin, K., Cohen, J. B., Li, D. & Kim, J. Quantifying uncertainty in ML-derived atmosphere remote sensing: Hourly surface NO2 estimation with GEMS. *Geophys. Res. Lett.* **51**, e2024GL110468. https://doi.org/10.1029/2024GL110468 (2024). _eprint: https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2024GL110468.
72. Singh, G. et al. Uncertainty quantification for probabilistic machine learning in earth observation using conformal prediction. *Sci. Rep.* **14**, 16166. https://doi.org/10.1038/s41598-024-65954-w (2024) (publisher: Nature Publishing Group).
73. Ferchichi, A., Chihaoui, M., Toujani, R., Ferchichi, A. & Hendaoui, F. Evidential uncertainty quantification with multiple deep learning architectures for spatiotemporal drought forecasting. *Neural Comput. Appl.* **37**, 8773–8797. https://doi.org/10.1007/s00521-025-11026-7 (2025).
74. Schreck, J. S. et al. Evidential deep learning: Enhancing predictive uncertainty estimation for earth system science applications. *Artif. Intell. Earth Syst.* **3**, 230093. https://doi.org/10.1175/AIES-D-23-0093.1 (2024).
75. Wang, L. et al. Quantification of model uncertainty and variability for landslide displacement prediction based on Monte Carlo simulation. *Gondwana Res.* **123**, 27–40. https://doi.org/10.1016/j.gr.2023.03.006 (2023).
76. Zhang, M., Fernndez-Torres, M.-A., Cohrs, K.-H. & Camps-Valls, G. Calibration and uncertainty quantification for deep learning-based drought detection. *Int. J. Appl. Earth Obs. Geoinf.* **140**, 104563. https://doi.org/10.1016/j.jag.2025.104563 (2025).
77. Yu, W. et al. A probabilistic approach to wildfire spread prediction using a denoising diffusion surrogate model. *EGUsphere*. 1–37. https://doi.org/10.5194/egusphere-2025-2836 (2025) (publisher: Copernicus GmbH).
78. Finn, T. S. et al. Generative diffusion for regional surrogate models from sea-ice simulations. *J. Adv. Model. Earth Syst.* **16**, e2024MS004395. https://doi.org/10.1029/2024MS004395 (2024). _eprint: https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2024MS004395.
79. Tu, S. et al. Satellite observations guided diffusion model for accurate meteorological states at arbitrary resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 28071–28080 (2025).
80. Sinha, S. et al. Variational autoencoder anomaly-detection of Avalanche deposits in satellite SAR imagery. In *Proceedings of the 10th International Conference on Climate Informatics*, CI2020. 113–119. https://doi.org/10.1145/3429309.3429326 (Association for Computing Machinery, 2021).
81. Yadav, R., Nascetti, A., Azizpour, H. & Ban, Y. Unsupervised flood detection on SAR time series using variational autoencoder. *Int. J. Appl. Earth Obs. Geoinf.* **126**, 103635. https://doi.org/10.1016/j.jag.2023.103635 (2024).
82. Lazaro-Gredilla, M., Titsias, M. K., Verrelst, J. & Camps-Valls, G. Retrieval of biophysical parameters with heteroscedastic Gaussian processes. *IEEE Geosci. Remote Sens. Lett.* **11**, 838–842. https://doi.org/10.1109/LGRS.2013.2279695 (2014).
83. Li, Y., Zhang, Y. & Zhu, Z. Learning deep networks under noisy labels for remote sensing image scene classification. In *IGARSS 2019 - 2019 IEEE International Geoscience and Remote Sensing Symposium*. 3025–3028. https://doi.org/10.1109/IGARSS.2019.8900497 (2019) (ISSN: 2153-7003).
84. Jiang, J., Ma, J., Wang, Z., Chen, C. & Liu, X. Hyperspectral image classification in the presence of noisy labels. In *IEEE Transactions on Geoscience and Remote Sensing*. Vol. 57. 851–865. https://doi.org/10.1109/TGRS.2018.2861992 (2019) (conference name: IEEE Transactions on Geoscience and Remote Sensing).
85. Li, P. et al. An improved categorical cross entropy for remote sensing image classification based on noisy labels. *Expert Syst. Appl.* **205**, 117296. https://doi.org/10.1016/j.eswa.2022.117296 (2022).
86. Xu, Y. et al. Dual-channel residual network for hyperspectral image classification with noisy labels. *IEEE Trans. Geosci. Remote Sens.* **60**, 1–11. https://doi.org/10.1109/TGRS.2021.3057689 (2022).
87. Cortés-Andrés, J., Fernández-Torres, M.-A. & Camps-Valls, G. Deep learning with noisy labels for spatiotemporal drought detection. *IEEE Trans. Geosci. Remote Sens.* **62**, 1–13. https://doi.org/10.1109/TGRS.2024.3504340 (2024).
88. Li, Q., Chen, Y. & Ghamisi, P. Complementary learning-based scene classification of remote sensing images with noisy labels. *IEEE Geosci. Remote Sens. Lett.* **19**, 1–5. https://doi.org/10.1109/LGRS.2021.3112960 (2022).
89. Aksoy, A. K., Ravanbakhsh, M. & Demir, B. Multi-label noise robust collaborative learning for remote sensing image classification. In *IEEE Transactions on Neural Networks and Learning Systems*. 1–14. https://doi.org/10.1109/TNNLS.2022.3209992 (2022). Conference name: IEEE Transactions on Neural Networks and Learning Systems.
90. Mirpulatov, I., Illarionova, S., Shadrin, D. & Burnaev, E. Pseudo-labeling approach for land cover classification through remote sensing observations with noisy labels. *IEEE Access* **11**, 82570–82583. https://doi.org/10.1109/ACCESS.2023.3300967 (2023).
91. Snow, R., O'Connor, B., Jurafsky, D. & Ng, A. Cheap and fast - But is it good? Evaluating non-expert annotations for natural language tasks. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*. 254–263 (Association for Computational Linguistics, 2008).
92. Raykar, V. C. et al. Learning from crowds. *J. Mach. Learn. Res.* **11**, 1297–1322 (2010).
93. Prapas, I. et al. Deep Learning Methods for Daily Wildfire Danger Forecasting (2021). ArXiv:2111.02736 [cs].
94. Sumbul, G., Charfuelan, M., Demir, B. & Markl, V. Bigearthnet: A large-scale benchmark archive for remote sensing image understanding. In *IGARSS 2019 - 2019 IEEE International Geoscience and Remote Sensing Symposium*. 5901–5904. https://doi.org/10.1109/IGARSS.2019.8900532 (IEEE, 2019).
95. Büttner, G. CORINE land cover and land cover change products. In *Land Use and Land Cover Mapping in Europe: Practices & Trends, Remote Sensing and Digital Image Processing* (Manakos, I. & Braun, M. eds.). 55–74. https://doi.org/10.1007/978-94-007-7969-3_5 (Springer, 2014).
96. He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778 (2016).
97. Zhou, Z., Siddiquee, M. M. R., Tajbakhsh, N. & Liang, J. UNet++: A Nested U-Net Architecture for Medical Image Segmentation (2018). arXiv:1807.10165 [cs, eess, stat].
98. Huang, G., Liu, Z., Van Der Maaten, L. & Weinberger, K. Q. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 4700–4708 (2017).
99. Hochreiter, S. & Schmidhuber, J. Long short-term memory. *Neural Comput.* **9**, 1735–80. https://doi.org/10.1162/neco.1997.9.8.1735 (1997).
100. Mukhoti, J. & Gal, Y. Evaluating Bayesian deep learning methods for semantic segmentation. *arXiv preprint* arXiv:1811.12709 (2019).
101. Haynes, K., Lagerquist, R., McGraw, M., Musgrave, K. & Ebert-Uphoff, I. Creating and evaluating uncertainty estimates with neural networks for environmental-science applications. *Artif. Intell. Earth Syst.* **2**, 220061 (2023).

102. Ståhl, N., Falkman, G., Karlsson, A. & Mathiason, G. Evaluation of uncertainty quantification in deep learning. In *Information Processing and Management of Uncertainty in Knowledge-Based Systems* (Lesot, M.-J. et al. eds.). 556–568. https://doi.org/10.1007/978-3-030-50146-4_41 (Springer, 2020).
103. Palmer, S. C., Kutser, T. & Hunter, P. D. Remote sensing of inland waters: Challenges, progress and future directions. *Remote Sens. Environ.* **157**, 1–8. https://doi.org/10.1016/j.rse.2014.09.021 (2015). Special issue: Remote Sensing of Inland Waters.
104. Giannaros, T. M. et al. Meteorological analysis of the 2021 extreme wildfires in Greece: Lessons learned and implications for early warning of the potential for pyroconvection. *Atmosphere* **13**. https://doi.org/10.3390/atmos13030475 (2022).
105. Kondylatos, S., Prapas, I., Camps-Valls, G. & Papoutsis, I. Mesogeos: A multi-purpose dataset for data-driven wildfire modeling in the Mediterranean. *Adv. Neural Inf. Process. Syst.* **36**, 50661–50676 (2023).
106. Farahani, A., Voghoei, S., Rasheed, K. & Arabnia, H. R. A brief review of domain adaptation. In *Advances in Data Science and Information Engineering* (Stahlbock, R. et al. eds.). 877–894 (Springer, 2021).
107. Settles, B. Active learning literature survey. In Technical Report, University of Wisconsin-Madison Department of Computer Sciences (2009). Accepted 15 Mar 2012.
108. Engelen, J. E. & Hoos, H. H. A survey on semi-supervised learning. *Mach. Learn.* **109**, 373–440. https://doi.org/10.1007/s10994-019-05855-6 (2020).
109. Rolf, E., Klemmer, K., Robinson, C. & Kerner, H. Mission critical–satellite data is a distinct modality in machine learning. *arXiv preprint* arXiv:2402.01444 (2024).

Acknowledgements

This work has received funding from the European Union's Horizon 2020 Research and Innovation under Grant Agreement no. 101036926 (TREEADS) and the European Union's Horizon Europe WIDERA Coordination and Support Actions under Grant Agreement no.101159723 (MeDiTwin).

Author contributions

S.K.: Conceptualization, Software, Methodology, Visualization, Writing-review & editing, Writing - original draft, Investigation, Data Curation N.I.B.: Software, Investigation, Data Curation I.P.: Software, Investigation, Data Curation, A.Z. Software, Investigation, Data Curation, G.C.-V.: Writing - Review & Editing, Supervision, Conceptualization, I.P.: Writing - Review & Editing, Supervision, Conceptualization, Resources, Project administration, Funding acquisition.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to S.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025